

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 13 · Integration

With Answers

Questions with full worked solutions

- 82 pages
- Past papers up to June 2026

Integration

Your classified route

33

questions

249

marks

4

families

10

variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Integration

Families and variants

F01 • Algebraic preprocessing and indefinite integration

V01 Direct sum of positive, negative and fractional powers

V02 Expand a product then integrate

V03 Expand or divide a radical or rational quotient

F02 • Recovering a curve from its first derivative

V01 Integrate f' and use one point

V02 Find parameter from gradient or line, then recover the curve

V03 Recover the curve and construct a tangent or normal

F03 • Recovering from second-derivative information

V01 Integrate twice using gradient and point conditions

V02 Use a second-derivative condition to find a parameter

F04 • Multi-constraint integration synthesis

V01 Determine multiple coefficients before recovering f

V02 Recovery plus a transformation or extra derivative demand

Algebraic preprocessing and indefinite integration

WHAT THIS FAMILY ASKS

Expand, divide or rewrite an integrand as a sum of powers, integrate term by term and include the arbitrary constant.

VARIANT ROUTE

V01 Direct sum of positive, negative and fractional powers

V02 Expand a product then integrate

V03 Expand or divide a radical or rational quotient

14 Complete Classified questions

68 total marks

1. Find, in simplest form,

$$\int \left(\frac{8x^3}{3} - \frac{1}{2\sqrt{x}} - 5 \right) dx$$

(4)

(Total 4 marks)

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Question 1

WMA11/01 January 2020

T13 Integration

F01 Algebraic preprocessing
and indefinite integrationV01 Direct sum of positive,
negative and fractional powers

1

Rewrite the integrand using powers

$$\frac{8x^3}{3} - \frac{1}{2\sqrt{x}} - 5 = \frac{8}{3}x^3 - \frac{1}{2}x^{-1/2} - 5.$$

2

Integrate term by term

$$\int \frac{8}{3}x^3 dx = \frac{2}{3}x^4, \int -\frac{1}{2}x^{-1/2} dx = -x^{1/2}, \text{ and } \int -5 dx = -5x.$$

3

Include the constant

$$\text{Therefore } \int \left(\frac{8x^3}{3} - \frac{1}{2\sqrt{x}} - 5 \right) dx = \frac{2}{3}x^4 - \sqrt{x} - 5x + c.$$

FINAL ANSWER

$$\frac{2}{3}x^4 - \sqrt{x} - 5x + c.$$

1. Find

$$\int 12x^3 + \frac{1}{6\sqrt{x}} - \frac{3}{2x^4} dx$$

giving each term in simplest form.

(5)

(Total 5 marks)

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Question 2

WMA11/01 October 2021

T13 Integration

F01 Algebraic preprocessing
and indefinite integrationV01 Direct sum of positive,
negative and fractional powers

1

Rewrite with powers

$$12x^3 + \frac{1}{6\sqrt{x}} - \frac{3}{2x^4} = 12x^3 + \frac{1}{6}x^{-1/2} - \frac{3}{2}x^{-4}.$$

2

Integrate term by term

$$\int 12x^3 dx = 3x^4, \int \frac{1}{6}x^{-1/2} dx = \frac{1}{3}x^{1/2}, \text{ and } \int -\frac{3}{2}x^{-4} dx = \frac{1}{2}x^{-3}.$$

3

Write in simplest form

$$\text{Therefore the integral is } 3x^4 + \frac{\sqrt{x}}{3} + \frac{1}{2x^3} + c.$$

FINAL ANSWER

$$3x^4 + \frac{\sqrt{x}}{3} + \frac{1}{2x^3} + c.$$

1. Find

$$\int \left(\frac{8x^3}{5} - \frac{2}{3x^4} - 1 \right) dx$$

giving each term in simplest form.

(4)

(Total 4 marks)

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Question 3

WMA11/01 January 2022

T13 Integration

F01 Algebraic preprocessing
and indefinite integrationV01 Direct sum of positive,
negative and fractional powers

1

Rewrite using powers

$$\frac{8x^3}{5} - \frac{2}{3x^4} - 1 = \frac{8}{5}x^3 - \frac{2}{3}x^{-4} - 1.$$

2

Integrate term by term

$$\int \frac{8}{5}x^3 dx = \frac{2}{5}x^4, \int -\frac{2}{3}x^{-4} dx = \frac{2}{9}x^{-3}, \text{ and } \int -1 dx = -x.$$

3

Write in simplest form

$$\text{Therefore the integral is } \frac{2x^4}{5} + \frac{2}{9x^3} - x + c.$$

FINAL ANSWER

$$\frac{2x^4}{5} + \frac{2}{9x^3} - x + c.$$

1. Find

$$\int \left(10x^5 + 6x^3 - \frac{3}{x^2} \right) dx$$

giving your answer in simplest form.

(4)

(Total 4 marks)

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Question 4

WMA11/01 May/June 2022

T13 Integration

F01 Algebraic preprocessing
and indefinite integrationV01 Direct sum of positive,
negative and fractional powers

1

Rewrite the negative power

$$\int \left(10x^5 + 6x^3 - \frac{3}{x^2} \right) dx = \int (10x^5 + 6x^3 - 3x^{-2}) dx.$$

2

Integrate term by term

$$\int 10x^5 dx = \frac{10x^6}{6} = \frac{5}{3}x^6, \int 6x^3 dx = \frac{6x^4}{4} = \frac{3}{2}x^4, \text{ and } \int -3x^{-2} dx = 3x^{-1}.$$

3

Add the constant

$$\text{Therefore the integral is } \frac{5}{3}x^6 + \frac{3}{2}x^4 + \frac{3}{x} + c.$$

FINAL ANSWER

$$\frac{5}{3}x^6 + \frac{3}{2}x^4 + \frac{3}{x} + c.$$

1. Find

$$\int \left(10x^4 - \frac{3}{2x^3} - 7 \right) dx$$

giving each term in simplest form.

(3)

(Total for Question 1 is 3 marks)

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Question 5

WMA11/01 May/June 2024

T13 Integration

F01 Algebraic preprocessing
and indefinite integrationV01 Direct sum of positive,
negative and fractional powers

1

Rewrite the fraction

$$10x^4 - \frac{3}{2x^3} - 7 = 10x^4 - \frac{3}{2}x^{-3} - 7.$$

2

Integrate term by term

$$\int 10x^4 dx = 2x^5, \int -\frac{3}{2}x^{-3} dx = \frac{3}{4}x^{-2}, \text{ and } \int -7 dx = -7x.$$

3

Write in simplest form

$$2x^5 + \frac{3}{4x^2} - 7x + c.$$

FINAL ANSWER

$$2x^5 + \frac{3}{4x^2} - 7x + c.$$

1. Find

$$\int \left(8x^3 - 6\sqrt{x} - \frac{2}{5x^3} \right) dx$$

giving your answer in simplest form.

(4)

(Total for Question 1 is 4 marks)

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Question 6

WMA11/01 January 2025

T13 Integration

F01 Algebraic preprocessing
and indefinite integrationV01 Direct sum of positive,
negative and fractional powers

1

Rewrite with powers

$$8x^3 - 6\sqrt{x} - \frac{2}{5x^3} = 8x^3 - 6x^{1/2} - \frac{2}{5}x^{-3}.$$

2

Integrate term by term

$$\int 8x^3 dx = 2x^4, \int -6x^{1/2} dx = -4x^{3/2}, \text{ and } \int -\frac{2}{5}x^{-3} dx = \frac{1}{5}x^{-2}.$$

3

Write in simplest form

$$2x^4 - 4x^{3/2} + \frac{1}{5x^2} + c.$$

FINAL ANSWER

$$2x^4 - 4x^{3/2} + \frac{1}{5x^2} + c.$$

1. $y = 4x^3 + \frac{3}{2\sqrt{x}} - 8 \quad x > 0$

Find, in simplest form,

(a) $\frac{dy}{dx}$ (3)

(b) $\int y \, dx$ (3)

(Total for Question 1 is 6 marks)

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Question 7

WMA11/01 January 2026

T13 Integration

F01 Algebraic preprocessing
and indefinite integrationV01 Direct sum of positive,
negative and fractional powers

1

Rewrite with powers

$$y = 4x^3 + \frac{3}{2}x^{-1/2} - 8.$$

2

Differentiate

$$\frac{dy}{dx} = 12x^2 + \frac{3}{2} \left(-\frac{1}{2} \right) x^{-3/2}.$$

3

Simplify derivative

$$\frac{dy}{dx} = 12x^2 - \frac{3}{4}x^{-3/2} = 12x^2 - \frac{3}{4x^{3/2}}.$$

4

Integrate

$$\int y \, dx = \int \left(4x^3 + \frac{3}{2}x^{-1/2} - 8 \right) dx.$$

5

Simplify integral

$$\int y \, dx = x^4 + 3x^{1/2} - 8x + c = x^4 + 3\sqrt{x} - 8x + c.$$

FINAL ANSWER

$$(a) \frac{dy}{dx} = 12x^2 - \frac{3}{4x^{3/2}}. \quad (b) \int y \, dx = x^4 + 3\sqrt{x} - 8x + c.$$

1. Find

$$\int (2x - 5)(3x + 2)(2x + 5)dx$$

writing your answer in simplest form.

(5)

(Total for Question 1 is 5 marks)

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Question 8

WMA11/01 January 2024

T13 Integration

F01 Algebraic preprocessing
and indefinite integrationV02 Expand a product then in-
tegrate

1

Expand two brackets first

$$(2x - 5)(3x + 2) = 6x^2 - 11x - 10.$$

2

Multiply by the third bracket

$$(6x^2 - 11x - 10)(2x + 5) = 12x^3 + 8x^2 - 75x - 50.$$

3

Integrate term by term

$$\int (12x^3 + 8x^2 - 75x - 50) dx = 3x^4 + \frac{8}{3}x^3 - \frac{75}{2}x^2 - 50x + c.$$

FINAL ANSWER

$$3x^4 + \frac{8}{3}x^3 - \frac{75}{2}x^2 - 50x + c.$$

2. Find

$$\int (x-3)^2 (2x+5) \, dx$$

writing the answer in simplest form.

(5)

(Total for Question 2 is 5 marks)

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Question 9

WMA11/01 October 2025

T13 Integration

F01 Algebraic preprocessing
and indefinite integrationV02 Expand a product then in-
tegrate

1

Expand the square

$$(x - 3)^2 = x^2 - 6x + 9.$$

2

Multiply out

$$(x^2 - 6x + 9)(2x + 5) = 2x^3 + 5x^2 - 12x^2 - 30x + 18x + 45.$$

3

Collect terms

$$(x - 3)^2(2x + 5) = 2x^3 - 7x^2 - 12x + 45.$$

4

Integrate term by term

$$\int (2x^3 - 7x^2 - 12x + 45)dx = \frac{1}{2}x^4 - \frac{7}{3}x^3 - 6x^2 + 45x + c.$$

FINAL ANSWER

$$\frac{1}{2}x^4 - \frac{7}{3}x^3 - 6x^2 + 45x + c.$$

1. Find

$$\int \left(3x + \frac{2}{x} \right)^2 dx$$

giving the answer in simplest form.

(5)

(Total for Question 1 is 5 marks)

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Question 10

WMA11/01 May/June 2026

T13 Integration

F01 Algebraic preprocessing
and indefinite integrationV02 Expand a product then in-
tegrate

1

Expand the square

$$\left(3x + \frac{2}{x}\right)^2 = 9x^2 + 12 + \frac{4}{x^2} = 9x^2 + 12 + 4x^{-2}.$$

2

Integrate term by term

$$\int (9x^2 + 12 + 4x^{-2}) \, dx = 3x^3 + 12x - 4x^{-1} + C.$$

3

Write the result in simplest form

Since $x^{-1} = \frac{1}{x}$, the integral is $3x^3 + 12x - \frac{4}{x} + C$.

FINAL ANSWER

$$3x^3 + 12x - \frac{4}{x} + C$$

4. Find

$$\int \frac{(3\sqrt{x} + 2)(x - 5)}{4\sqrt{x}} dx$$

writing each term in simplest form.

(6)

(Total 6 marks)

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Question 11

WMA11/01 May/June 2021

T13 Integration

F01 Algebraic preprocessing
and indefinite integration

V03 Expand or divide a radical or rational quotient

1

Expand the numerator

$$(3\sqrt{x} + 2)(x - 5) = 3x\sqrt{x} - 15\sqrt{x} + 2x - 10.$$

2

Divide by $4\sqrt{x}$

$$\frac{(3\sqrt{x} + 2)(x - 5)}{4\sqrt{x}} = \frac{3x}{4} - \frac{15}{4} + \frac{1}{2}x^{1/2} - \frac{5}{2}x^{-1/2}.$$

3

Integrate term by term

$$\int \left(\frac{3x}{4} - \frac{15}{4} + \frac{1}{2}x^{1/2} - \frac{5}{2}x^{-1/2} \right) dx = \frac{3x^2}{8} - \frac{15x}{4} + \frac{1}{3}x^{3/2} - 5x^{1/2} + c.$$

4

Write each term simply

So the integral is $\frac{3x^2}{8} - \frac{15x}{4} + \frac{x\sqrt{x}}{3} - 5\sqrt{x} + c.$

FINAL ANSWER

$$\frac{3x^2}{8} - \frac{15x}{4} + \frac{x\sqrt{x}}{3} - 5\sqrt{x} + c.$$

2. Given that

$$(x - 5)(2x + 1)(x + 3) \equiv ax^3 + bx^2 - 32x - 15$$

where a and b are constants,

(a) find the value of a and the value of b .

(2)

(b) Hence find

$$\int \frac{(x - 5)(2x + 1)(x + 3)}{5\sqrt{x}} dx$$

writing each term in simplest form.

(5)

(Total for Question 2 is 7 marks)

Question 12

WMA11/01 October 2022

T13 Integration

F01 Algebraic preprocessing
and indefinite integration

V03 Expand or divide a radical or rational quotient

1

Expand the identity $(2x + 1)(x + 3) = 2x^2 + 7x + 3$, so $(x - 5)(2x^2 + 7x + 3) = 2x^3 - 3x^2 - 32x - 15$.

2

Read off a and bComparing with $ax^3 + bx^2 - 32x - 15$ gives $a = 2$ and $b = -3$.

3

Rewrite the integrand

$$\frac{(x - 5)(2x + 1)(x + 3)}{5\sqrt{x}} = \frac{1}{5}(2x^{5/2} - 3x^{3/2} - 32x^{1/2} - 15x^{-1/2}).$$

4

Integrate term by term

$$\int \frac{1}{5}(2x^{5/2})dx = \frac{4}{35}x^{7/2} \text{ and } \int \frac{1}{5}(-3x^{3/2})dx = -\frac{6}{25}x^{5/2}.$$

5

Complete the integral

$$\int \frac{1}{5}(-32x^{1/2})dx = -\frac{64}{15}x^{3/2} \text{ and } \int \frac{1}{5}(-15x^{-1/2})dx = -6x^{1/2}, \text{ then add } c.$$

FINAL ANSWER

$$(a) a = 2, b = -3. \quad (b) \frac{4}{35}x^{7/2} - \frac{6}{25}x^{5/2} - \frac{64}{15}x^{3/2} - 6\sqrt{x} + c.$$

3. Find

$$\int \frac{4x^5 + 3}{2x^2} dx$$

giving your answer in simplest form.

(5)

(Total for Question 3 is 5 marks)

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Question 13

WMA11/01 January 2023

T13 Integration

F01 Algebraic preprocessing
and indefinite integration

V03 Expand or divide a radical or rational quotient

1

Split the fraction

$$\frac{4x^5 + 3}{2x^2} = 2x^3 + \frac{3}{2}x^{-2}.$$

2

Integrate the first term

$$\int 2x^3 dx = \frac{x^4}{2}.$$

3

Integrate the second term

$$\int \frac{3}{2}x^{-2} dx = -\frac{3}{2}x^{-1}.$$

4

Write in simplest form

The integral is $\frac{x^4}{2} - \frac{3}{2x} + c.$

FINAL ANSWER

$$\frac{x^4}{2} - \frac{3}{2x} + c.$$

3. Find

$$\int \frac{(x+3)^2}{3\sqrt{x}} dx$$

writing your answer in simplest form.

(5)

(Total for Question 3 is 5 marks)

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Question 14

WMA11/01 May/June 2025

T13 Integration

F01 Algebraic preprocessing
and indefinite integration

V03 Expand or divide a radical or rational quotient

1

Expand the numerator

$$(x + 3)^2 = x^2 + 6x + 9.$$

2

Divide by the denominator

$$\frac{(x + 3)^2}{3\sqrt{x}} = \frac{x^2 + 6x + 9}{3x^{1/2}} = \frac{1}{3}x^{3/2} + 2x^{1/2} + 3x^{-1/2}.$$

3

Integrate term by term

$$\int \left(\frac{1}{3}x^{3/2} + 2x^{1/2} + 3x^{-1/2} \right) dx = \frac{1}{3} \cdot \frac{2}{5}x^{5/2} + 2 \cdot \frac{2}{3}x^{3/2} + 3 \cdot 2x^{1/2} + c.$$

4

Simplify

$$\frac{2}{15}x^{5/2} + \frac{4}{3}x^{3/2} + 6x^{1/2} + c.$$

FINAL ANSWER

$$\frac{2}{15}x^{5/2} + \frac{4}{3}x^{3/2} + 6x^{1/2} + c.$$

Recovering a curve from its first derivative

WHAT THIS FAMILY ASKS

Integrate f' , determine the constant from a point and connect the derivative to tangent or normal information when present.

VARIANT ROUTE

V02 Find parameter from gradient or line, then recover the curve

V03 Recover the curve and construct a tangent or normal

11 Complete Classified questions

107 total marks

8. A curve C has equation $y = f(x)$.

The point P with x coordinate 3 lies on C

Given

- $f'(x) = 4x^2 + kx + 3$ where k is a constant
- the normal to C at P has equation $y = -\frac{1}{24}x + 5$

(a) show that $k = -5$

(3)

(b) Hence find $f(x)$.

(4)

(Total for Question 8 is 7 marks)

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Question 15

WMA11/01 October 2024

T13 Integration

F02 Recovering a curve from its first derivative

V02 Find parameter from gradient or line, then recover the curve

1

Find the y-coordinate of P

The normal is $y = -\frac{1}{24}x + 5$. At $x = 3$, $y = -\frac{3}{24} + 5 = \frac{39}{8}$.

2

Use the normal gradient

The normal gradient is $-\frac{1}{24}$, so the tangent gradient is 24.

3

Use f prime at x equals 3

$$f'(3) = 4(3)^2 + 3k + 3 = 39 + 3k.$$

4

Find k

$$39 + 3k = 24, \text{ so } k = -5.$$

5

Integrate f prime

$$\text{With } k = -5, f'(x) = 4x^2 - 5x + 3, \text{ so } f(x) = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 3x + c.$$

6

Use point P

$$\frac{39}{8} = \frac{4}{3}(27) - \frac{5}{2}(9) + 3(3) + c = \frac{45}{2} + c.$$

7

Find c

$$c = \frac{39}{8} - \frac{45}{2} = -\frac{141}{8}.$$

FINAL ANSWER

$$k = -5 \text{ and } f(x) = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 3x - \frac{141}{8}.$$

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8.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

A curve has equation $y = f(x)$, $x > 0$

The point $P(4, 12)$ lies on the curve.

Given that

- $f'(x) = 3\sqrt{x} + kx^2$ where k is a constant
- the equation of the tangent to the curve at P has equation $y = 10x + c$ where c is a constant

(a) (i) show that $k = \frac{1}{4}$

(ii) find the value of c

(4)

(b) Hence find the value of $f''(x)$ at P .

(3)

(c) Find $f(x)$.

(4)

(Total for Question 8 is 11 marks)

Question 16

WMA11/01 May/June 2025

T13 Integration

F02 Recovering a curve from its first derivative

V02 Find parameter from gradient or line, then recover the curve

1

Use the tangent gradient**The tangent at P has equation $y = 10x + c$, so the gradient at P is 10.**

2

Show k

$$f'(4) = 3\sqrt{4} + k(4)^2 = 6 + 16k = 10, \text{ so } 16k = 4 \text{ and } k = \frac{1}{4}.$$

3

Find c**Substitute $P(4, 12)$ into $y = 10x + c$: $12 = 40 + c$, so $c = -28$.**

4

Differentiate f prime

$$\text{With } k = \frac{1}{4}, f'(x) = 3x^{1/2} + \frac{1}{4}x^2.$$

5

Find f double prime

$$f''(x) = \frac{3}{2}x^{-1/2} + \frac{1}{2}x.$$

6

Evaluate at P

$$f''(4) = \frac{3}{2} \cdot \frac{1}{2} + 2 = \frac{3}{4} + 2 = \frac{11}{4}.$$

7

Integrate f prime

$$f(x) = \int \left(3x^{1/2} + \frac{1}{4}x^2 \right) dx = 2x^{3/2} + \frac{1}{12}x^3 + d.$$

8

Use P to find d

$$12 = 2(4^{3/2}) + \frac{1}{12}(4^3) + d = 16 + \frac{16}{3} + d.$$

9

State f(x)

$$d = -\frac{28}{3}, \text{ so } f(x) = 2x^{3/2} + \frac{1}{12}x^3 - \frac{28}{3}.$$

FINAL ANSWER

$$(a)(i) k = \frac{1}{4}. \quad (a)(ii) c = -28. \quad (b) f''(P) = \frac{11}{4}. \quad (c) f(x) = 2x^{3/2} + \frac{1}{12}x^3 - \frac{28}{3}.$$

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8.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

A curve has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = 2x + \frac{8}{x^2} + k$, where k is a constant
- the equation of the tangent to the curve at $x = \sqrt{2}$ is $y = 5x - 3\sqrt{2}$

(a) find the exact value of k .

(2)

(b) Find an equation of the normal to the curve at $x = \sqrt{2}$

(2)

(c) Find $f(x)$, writing your answer in simplest form.

(4)

(Total for Question 8 is 8 marks)

Question 17

WMA11/01 October 2025

T13 Integration

F02 Recovering a curve from its first derivative

V02 Find parameter from gradient or line, then recover the curve

1

Use tangent gradientThe tangent has equation $y = 5x - 3\sqrt{2}$, so the gradient at $x = \sqrt{2}$ is 5.

2

Find k

$$f'(\sqrt{2}) = 2\sqrt{2} + \frac{8}{(\sqrt{2})^2} + k = 2\sqrt{2} + 4 + k = 5, \text{ so } k = 1 - 2\sqrt{2}.$$

3

Find the point on the curveAt $x = \sqrt{2}$, the tangent gives $y = 5\sqrt{2} - 3\sqrt{2} = 2\sqrt{2}$.

4

Normal gradientThe tangent gradient is 5, so the normal gradient is $-\frac{1}{5}$.

5

Equation of the normal

$$y - 2\sqrt{2} = -\frac{1}{5}(x - \sqrt{2}), \text{ hence } y = -\frac{1}{5}x + \frac{11\sqrt{2}}{5}.$$

6

Integrate f prime

$$f(x) = \int \left(2x + \frac{8}{x^2} + k \right) dx = x^2 - \frac{8}{x} + kx + c.$$

7

Substitute k

$$f(x) = x^2 - \frac{8}{x} + (1 - 2\sqrt{2})x + c.$$

8

Use the point

$$2\sqrt{2} = 2 - \frac{8}{\sqrt{2}} + (1 - 2\sqrt{2})\sqrt{2} + c = -2 - 3\sqrt{2} + c.$$

9

Find c

$$c = 2 + 5\sqrt{2}, \text{ so } f(x) = x^2 - \frac{8}{x} + (1 - 2\sqrt{2})x + 2 + 5\sqrt{2}.$$

FINAL ANSWER

$$(a) k = 1 - 2\sqrt{2}. \quad (b) y = -\frac{1}{5}x + \frac{11\sqrt{2}}{5}. \quad (c) f(x) = x^2 - \frac{8}{x} + (1 - 2\sqrt{2})x + 2 + 5\sqrt{2}.$$

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8.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

A curve C has equation $y = f(x)$, $x > 0$

A point P lies on C .

Given that

- $f'(x) = 4x^2 + \frac{6}{x^2} - 4$
- the equation of the tangent to C at P is $y = 10x - 6\sqrt{3}$

(a) (i) verify that $\sqrt{3}$ is a possible x coordinate of P ,

(ii) find, using algebra, the other possible x coordinate of P .

(6)

Given that the x coordinate of P is $\sqrt{3}$

(b) find an equation of the normal to C at P .

(2)

(c) Find $f(x)$, writing your answer in simplest form.
You must show each stage of your working.

(4)

(Total for Question 8 is 12 marks)

Question 18

WMA11/01 January 2026

T13 Integration

F02 Recovering a curve from its first derivative

V02 Find parameter from gradient or line, then recover the curve

1

Verify x equals root 3

$f'(\sqrt{3}) = 4(\sqrt{3})^2 + \frac{6}{(\sqrt{3})^2} - 4 = 12 + 2 - 4 = 10$, which matches the tangent gradient.

2

Find all possible x-coordinates

Set $4x^2 + \frac{6}{x^2} - 4 = 10$.

3

Form a quadratic in x squared

Multiplying by x^2 gives $4x^4 - 14x^2 + 6 = 0$, so $2x^4 - 7x^2 + 3 = 0$.

4

Factor

$2x^4 - 7x^2 + 3 = (2x^2 - 1)(x^2 - 3) = 0$.

5

Use x greater than zero

$x = \sqrt{3}$ or $x = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$. The other possible x-coordinate is $\frac{\sqrt{2}}{2}$.

6

Find the point P

Given $x_P = \sqrt{3}$, the tangent gives $y_P = 10\sqrt{3} - 6\sqrt{3} = 4\sqrt{3}$.

7

Normal equation

The normal gradient is $-\frac{1}{10}$, so $y - 4\sqrt{3} = -\frac{1}{10}(x - \sqrt{3})$.

8

Integrate f prime

$$f(x) = \int \left(4x^2 + \frac{6}{x^2} - 4 \right) dx = \frac{4}{3}x^3 - \frac{6}{x} - 4x + c.$$

9

Use P to find c

$$4\sqrt{3} = \frac{4}{3}(\sqrt{3})^3 - \frac{6}{\sqrt{3}} - 4\sqrt{3} + c = 4\sqrt{3} - 2\sqrt{3} - 4\sqrt{3} + c.$$

10

State f(x)

$$c = 6\sqrt{3}, \text{ so } f(x) = \frac{4}{3}x^3 - \frac{6}{x} - 4x + 6\sqrt{3}.$$

FINAL ANSWER

(a)(i) **verified.** (a)(ii) $x = \frac{\sqrt{2}}{2}$. (b) $y - 4\sqrt{3} = -\frac{1}{10}(x - \sqrt{3})$. (c) $f(x) = \frac{4}{3}x^3 - \frac{6}{x} - 4x + 6\sqrt{3}$.

8. A curve C has equation $y = f(x)$, $x > 0$

The point P with x coordinate $\frac{1}{4}$ lies on C .

Given that

- $f'(x) = 10 + \frac{5\sqrt{x} - 4}{2x^2}$
- the tangent to C at P has equation $y = kx + 2$, where k is a constant

(a) (i) find the value of k

(ii) find the equation of the normal to C at P .

Give the answer in the form $y = mx + c$, where m and c are constants.

(6)

(b) Find $f(x)$.

(5)

(Total for Question 8 is 11 marks)

Question 19

WMA11/01 May/June 2026

T13 Integration

F02 Recovering a curve from its first derivative

V02 Find parameter from gradient or line, then recover the curve

1

(a)(i) Evaluate the tangent gradientThe tangent gradient at P is $f'\left(\frac{1}{4}\right)$. Using

$$f'(x) = 10 + \frac{5\sqrt{x} - 4}{2x^2},$$

we get

$$f'\left(\frac{1}{4}\right) = 10 + \frac{5\left(\frac{1}{2}\right) - 4}{2\left(\frac{1}{4}\right)^2} = 10 + \frac{-\frac{3}{2}}{\frac{1}{8}} = -2.$$

Therefore $k = -2$.

2

(a)(ii) Find the point of contactThe tangent is $y = -2x + 2$. At $x = \frac{1}{4}$,

$$y = -2\left(\frac{1}{4}\right) + 2 = \frac{3}{2},$$

so $P = \left(\frac{1}{4}, \frac{3}{2}\right)$.

3

(a)(ii) Form the normalThe tangent gradient is -2 , so the normal gradient is $\frac{1}{2}$. Through P ,

$$y - \frac{3}{2} = \frac{1}{2}\left(x - \frac{1}{4}\right).$$

Hence

$$y = \frac{1}{2}x + \frac{11}{8}.$$

4

(b) Rewrite and integrate

First split the fractional term:

$$f'(x) = 10 + \frac{5}{2}x^{-3/2} - 2x^{-2}.$$

Integrating term by term gives

$$f(x) = 10x - 5x^{-1/2} + 2x^{-1} + c = 10x - \frac{5}{\sqrt{x}} + \frac{2}{x} + c.$$

5

(b) Use the point P to find the constantSince $P = \left(\frac{1}{4}, \frac{3}{2}\right)$ lies on the curve,

$$\frac{3}{2} = 10\left(\frac{1}{4}\right) - 5\left(\frac{1}{4}\right)^{-1/2} + 2\left(\frac{1}{4}\right)^{-1} + c.$$

Thus

$$\frac{3}{2} = \frac{5}{2} - 10 + 8 + c = \frac{1}{2} + c,$$

so $c = 1$.**FINAL ANSWER**

$$(a)(i) k = -2. \quad (a)(ii) y = \frac{1}{2}x + \frac{11}{8}. \quad (b) f(x) = 10x - \frac{5}{\sqrt{x}} + \frac{2}{x} + 1.$$

6. The curve C has equation $y = f(x)$, $x > 0$

Given that

- C passes through the point $P(8, 2)$

- $f'(x) = \frac{32}{3x^2} + 3 - 2(\sqrt[3]{x})$

- (a) find the equation of the tangent to C at P . Write your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

- (b) Find, in simplest form, $f(x)$.

(5)

(Total 8 marks)

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Question 20

WMA11/01 May/June 2021

T13 Integration

F02 Recovering a curve from its first derivative

V03 Recover the curve and construct a tangent or normal

1

Find the gradient at P

$$f'(x) = \frac{32}{3x^2} + 3 - 2\sqrt[3]{x}. \text{ At } x = 8, f'(8) = \frac{32}{3(64)} + 3 - 2(2) = \frac{1}{6} - 1 = -\frac{5}{6}.$$

2

Write the tangent equation

$$\text{Using } P(8, 2), y - 2 = -\frac{5}{6}(x - 8), \text{ hence } y = -\frac{5}{6}x + \frac{26}{3}.$$

3

Integrate $f'(x)$

$$f'(x) = \frac{32}{3}x^{-2} + 3 - 2x^{1/3}, \text{ so } f(x) = -\frac{32}{3x} + 3x - \frac{3}{2}x^{4/3} + c.$$

4

Use P(8,2) to find c

$$2 = -\frac{32}{24} + 24 - \frac{3}{2}(16) + c = -\frac{4}{3} + c, \text{ so } c = \frac{10}{3}.$$

5

Write $f(x)$

$$f(x) = -\frac{32}{3x} + 3x - \frac{3}{2}x^{4/3} + \frac{10}{3}.$$

FINAL ANSWER

$$(a) y = -\frac{5}{6}x + \frac{26}{3}. \quad (b) f(x) = -\frac{32}{3x} + 3x - \frac{3}{2}x^{4/3} + \frac{10}{3}.$$

6. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \frac{(x+3)^2}{x\sqrt{x}}$
- the point $P(4, 20)$ lies on C

(a) (i) find the value of the gradient at P

(ii) Hence find the equation of the tangent to C at P , giving your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

(b) Find $f(x)$, simplifying your answer.

(7)

(Total 11 marks)

Question 21

WMA11/01 January 2022

T13 Integration

F02 Recovering a curve from its first derivative

V03 Recover the curve and construct a tangent or normal

1

Simplify $f'(x)$

$$f'(x) = \frac{(x+3)^2}{x\sqrt{x}} = \frac{x^2 + 6x + 9}{x^{3/2}} = x^{1/2} + 6x^{-1/2} + 9x^{-3/2}.$$

2

Find the gradient at P

$$\text{At } x = 4, f'(4) = 2 + 3 + \frac{9}{8} = \frac{49}{8}.$$

3

Find the tangent equation

$$\text{Using } P(4, 20), y - 20 = \frac{49}{8}(x - 4). \text{ Multiplying out gives } 49x - 8y - 36 = 0.$$

4

Integrate $f'(x)$

$$f(x) = \int (x^{1/2} + 6x^{-1/2} + 9x^{-3/2}) dx = \frac{2}{3}x^{3/2} + 12x^{1/2} - 18x^{-1/2} + c.$$

5

Use P(4,20)

$$20 = \frac{2}{3}(8) + 12(2) - \frac{18}{2} + c = \frac{61}{3} + c, \text{ so } c = -\frac{1}{3}.$$

6

Write $f(x)$

$$f(x) = \frac{2}{3}x\sqrt{x} + 12\sqrt{x} - \frac{18}{\sqrt{x}} - \frac{1}{3}.$$

FINAL ANSWER

$$(a)(i) \frac{49}{8}. \quad (a)(ii) 49x - 8y - 36 = 0. \quad (b) f(x) = \frac{2}{3}x\sqrt{x} + 12\sqrt{x} - \frac{18}{\sqrt{x}} - \frac{1}{3}.$$

5.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = \frac{12}{\sqrt{x}} + \frac{x}{3} - 4$
- the point $P(9, 8)$ lies on C

(a) find, in simplest form, $f(x)$

(5)

The line l is the normal to C at P

(b) Find the coordinates of the point at which l crosses the y -axis.

(4)

(Total for Question 5 is 9 marks)

Question 22

WMA11/01 October 2022

T13 Integration

F02 Recovering a curve from its first derivative

V03 Recover the curve and construct a tangent or normal

1

Integrate $f'(x)$

$$f'(x) = 12x^{-1/2} + \frac{x}{3} - 4, \text{ so } f(x) = 24\sqrt{x} + \frac{x^2}{6} - 4x + c.$$

2

Use $P(9, 8)$

$$8 = 24\sqrt{9} + \frac{9^2}{6} - 4(9) + c = 72 + \frac{81}{6} - 36 + c.$$

3

Find c

$$72 + \frac{81}{6} - 36 = \frac{99}{2}, \text{ so } c = 8 - \frac{99}{2} = -\frac{83}{2}.$$

4

Find the tangent gradient at P

$$f'(9) = \frac{12}{3} + \frac{9}{3} - 4 = 3.$$

5

Find the normal gradient

The normal gradient is the negative reciprocal of 3, so it is $-\frac{1}{3}$.

6

Find the y-intercept

The normal through $(9, 8)$ is $y - 8 = -\frac{1}{3}(x - 9)$, so $y = -\frac{1}{3}x + 11$. It crosses the y-axis at $(0, 11)$.

FINAL ANSWER

$$(a) f(x) = 24\sqrt{x} + \frac{x^2}{6} - 4x - \frac{83}{2}. \quad (b) (0, 11).$$

8.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Find the equation of the tangent to the curve with equation

$$y = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

at the point $P(4, 12)$

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(5)

The curve with equation $y = f(x)$ also passes through the point $P(4, 12)$

Given that

$$f'(x) = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

(b) find $f(x)$ giving the coefficients in simplest form.

(5)

(Total for Question 8 is 10 marks)

Question 23

WMA11/01 May/June 2023

T13 Integration

F02 Recovering a curve from its first derivative

V03 Recover the curve and construct a tangent or normal

1

Differentiate the curve

For $y = \frac{1}{4}x^3 - 8x^{-1/2}$, $\frac{dy}{dx} = \frac{3}{4}x^2 + 4x^{-3/2}$.

2

Find the gradient at P

At $x = 4$, $\frac{dy}{dx} = \frac{3}{4}(16) + 4(4^{-3/2}) = 12 + \frac{1}{2} = \frac{25}{2}$.

3

Find the tangent

Using $P(4, 12)$, $y - 12 = \frac{25}{2}(x - 4)$.

4

Write in integer form

Multiplying by 2 gives $2y - 24 = 25x - 100$, hence $25x - 2y - 76 = 0$.

5

Integrate $f'(x)$

$f'(x) = \frac{1}{4}x^3 - 8x^{-1/2}$ gives $f(x) = \frac{x^4}{16} - 16\sqrt{x} + c$.

6

Use $P(4, 12)$

$12 = \frac{4^4}{16} - 16\sqrt{4} + c = 16 - 32 + c$, so $c = 28$.

FINAL ANSWER

(a) $25x - 2y - 76 = 0$. (b) $f(x) = \frac{x^4}{16} - 16\sqrt{x} + 28$.

7. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \frac{4x^2 + 10 - 7x^{\frac{1}{2}}}{4x^{\frac{1}{2}}}$

- the point $P(4, -1)$ lies on C

(a) (i) find the value of the gradient of C at P

(ii) Hence find the equation of the normal to C at P , giving your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

(b) Find $f(x)$.

(6)

(Total for Question 7 is 10 marks)

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Question 24

WMA11/01 October 2023

T13 Integration

F02 Recovering a curve from its first derivative

V03 Recover the curve and construct a tangent or normal

1

Simplify f' prime

$$f'(x) = \frac{4x^2 + 10 - 7x^{1/2}}{4x^{1/2}} = x^{3/2} + \frac{5}{2}x^{-1/2} - \frac{7}{4}.$$

2

Find the gradient at P

$$\text{At } x = 4, f'(4) = 4^{3/2} + \frac{5}{2}(4^{-1/2}) - \frac{7}{4} = 8 + \frac{5}{4} - \frac{7}{4} = \frac{15}{2}.$$

3

Find the normal gradient

The normal gradient is the negative reciprocal, so $m_n = -\frac{2}{15}$.

4

Use point P

$$\text{Using } P(4, -1), y + 1 = -\frac{2}{15}(x - 4).$$

5

Write the normal in integer form

$$15y + 15 = -2x + 8, \text{ hence } 2x + 15y + 7 = 0.$$

6

Integrate f' prime

$$f(x) = \int \left(x^{3/2} + \frac{5}{2}x^{-1/2} - \frac{7}{4} \right) dx = \frac{2}{5}x^{5/2} + 5x^{1/2} - \frac{7}{4}x + c.$$

7

Use P to find c

$$-1 = \frac{2}{5}(4^{5/2}) + 5(4^{1/2}) - \frac{7}{4}(4) + c = \frac{64}{5} + 10 - 7 + c, \text{ so } c = -\frac{84}{5}.$$

FINAL ANSWER

$$(a)(i) \frac{15}{2}. \quad (a)(ii) 2x + 15y + 7 = 0. \quad (b) f(x) = \frac{2}{5}x^{5/2} + 5x^{1/2} - \frac{7}{4}x - \frac{84}{5}.$$

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10. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = 6x - \frac{(2x-1)(3x+2)}{2\sqrt{x}}$

- the point $P(4, 12)$ lies on C

(a) find the equation of the normal to C at P , giving your answer in the form $y = mx + c$ where m and c are integers to be found,

(4)

(b) find $f(x)$, giving each term in simplest form.

(6)

(Total for Question 10 is 10 marks)

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Question 25

WMA11/01 May/June 2024

T13 Integration

F02 Recovering a curve from its first derivative

V03 Recover the curve and construct a tangent or normal

1

Expand the numerator

$$(2x - 1)(3x + 2) = 6x^2 + x - 2.$$

2

Simplify f prime

$$f'(x) = 6x - \frac{6x^2 + x - 2}{2\sqrt{x}} = 6x - 3x^{3/2} - \frac{1}{2}x^{1/2} + x^{-1/2}.$$

3

Find the tangent gradient at P

$$f'(4) = 24 - 3(8) - \frac{1}{2}(2) + \frac{1}{2} = -\frac{1}{2}.$$

4

Use the normal gradient

The normal gradient is the negative reciprocal of $-\frac{1}{2}$, so it is 2.

5

Find the normal equation

Through $P(4, 12)$, $y - 12 = 2(x - 4)$, hence $y = 2x + 4$.

6

Integrate f prime

$$f(x) = 3x^2 - \frac{6}{5}x^{5/2} - \frac{1}{3}x^{3/2} + 2x^{1/2} + c.$$

7

Use P to find c

$$12 = 3(4)^2 - \frac{6}{5}(4^{5/2}) - \frac{1}{3}(4^{3/2}) + 2(4^{1/2}) + c.$$

8

Simplify c

$$12 = 48 - \frac{192}{5} - \frac{8}{3} + 4 + c = \frac{164}{15} + c, \text{ so } c = \frac{16}{15}.$$

FINAL ANSWER

$$(a) y = 2x + 4. \quad (b) f(x) = 3x^2 - \frac{6}{5}x^{5/2} - \frac{1}{3}x^{3/2} + 2x^{1/2} + \frac{16}{15}.$$

DR ESLAM AHMED

Recovering from second-derivative information

WHAT THIS FAMILY ASKS

Use f'' either to reconstruct f through two integrations or to determine a parameter in f' before integrating once.

VARIANT ROUTE

V01 Integrate twice using gradient and point conditions

V02 Use a second-derivative condition to find a parameter

6 Complete Classified questions

53 total marks

11. A curve has equation $y = f(x)$.

The point $P\left(4, \frac{32}{3}\right)$ lies on the curve.

Given that

- $f''(x) = \frac{4}{\sqrt{x}} - 3$
- $f'(x) = 5$ at P

find

(a) the equation of the tangent to the curve at P , writing your answer in the form $y = mx + c$, where m and c are constants to be found,

(2)

(b) $f(x)$.

(8)

(Total 10 marks)

Question 26

WMA11/01 October 2019

T13 Integration

F03 Recovering from second-
derivative informationV01 Integrate twice using gra-
dient and point conditions

1

Use the given gradient for the tangentAt $P\left(4, \frac{32}{3}\right)$, $f'(4) = 5$, so the tangent has gradient 5.

2

Find the tangent equation $y - \frac{32}{3} = 5(x - 4)$. Hence $y = 5x - \frac{28}{3}$.

3

Integrate f'' to find f' $f''(x) = 4x^{-1/2} - 3$, so $f'(x) = 8x^{1/2} - 3x + C$.

4

Use $f'(4)=5$ $5 = 8(2) - 12 + C = 4 + C$, so $C = 1$. Therefore $f'(x) = 8\sqrt{x} - 3x + 1$.

5

Integrate again $f(x) = \frac{16}{3}x^{3/2} - \frac{3}{2}x^2 + x + D$.

6

Use the point on the curveSubstitute $x = 4$, $f(4) = \frac{32}{3}$: $\frac{32}{3} = \frac{16}{3}(8) - \frac{3}{2}(16) + 4 + D = \frac{68}{3} + D$, so $D = -12$.

FINAL ANSWER

(a) $y = 5x - \frac{28}{3}$. (b) $f(x) = \frac{16}{3}x^{3/2} - \frac{3}{2}x^2 + x - 12$.

11. A curve has equation $y = f(x)$, where

$$f''(x) = \frac{6}{\sqrt{x^3}} + x \quad x > 0$$

The point $P(4, -50)$ lies on the curve.

Given that $f'(x) = -4$ at P ,

- (a) find the equation of the normal at P , writing your answer in the form $y = mx + c$, where m and c are constants,

(3)

- (b) find $f(x)$.

(8)

(Total 11 marks)

Question 27

WMA11/01 January 2020

T13 Integration

F03 Recovering from second-
derivative informationV01 Integrate twice using gra-
dient and point conditions

1

Find the normal gradient

At P , $f'(4) = -4$, so the tangent gradient is -4 . The normal gradient is therefore $\frac{1}{4}$.

2

Find the normal equation

Using $P(4, -50)$: $y + 50 = \frac{1}{4}(x - 4)$, so $y = \frac{1}{4}x - 51$.

3

Integrate f'' to find f'

$f''(x) = 6x^{-3/2} + x$. Therefore $f'(x) = -12x^{-1/2} + \frac{1}{2}x^2 + k$.

4

Use $f'(4) = -4$

$-4 = -\frac{12}{2} + \frac{1}{2}(16) + k = -6 + 8 + k = 2 + k$, so $k = -6$.

5

Integrate again

$f(x) = -24x^{1/2} + \frac{x^3}{6} - 6x + d$.

6

Use the point on the curve

Substitute $P(4, -50)$: $-50 = -24(2) + \frac{64}{6} - 24 + d = -72 + \frac{32}{3} + d = -\frac{184}{3} + d$.
Hence $d = \frac{34}{3}$.

FINAL ANSWER

$$(a) y = \frac{1}{4}x - 51. \quad (b) f(x) = -24\sqrt{x} + \frac{x^3}{6} - 6x + \frac{34}{3}.$$

11. A curve C has equation $y = f(x)$, $x > 0$

Given that

- $f''(x) = 4x + \frac{1}{\sqrt{x}}$
- the point P has x coordinate 4 and lies on C
- the tangent to C at P has equation $y = 3x + 4$

(a) find an equation of the normal to C at P

(2)

(b) find $f(x)$, writing your answer in simplest form.

(6)

(Total for Question 11 is 8 marks)

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Question 28

WMA11/01 January 2023

T13 Integration

F03 Recovering from second-
derivative informationV01 Integrate twice using gra-
dient and point conditions

1

Find point P

Since P has x-coordinate 4 and lies on the tangent $y = 3x + 4$, $P = (4, 16)$.

2

Find the normal

The tangent gradient is 3, so the normal gradient is $-\frac{1}{3}$. Hence the normal is $y - 16 = -\frac{1}{3}(x - 4)$.

3

Integrate $f''(x)$

$f''(x) = 4x + x^{-1/2}$, so $f'(x) = 2x^2 + 2\sqrt{x} + A$.

4

Use the tangent gradient

At $x = 4$, $f'(4) = 3$. Thus $2(16) + 2(2) + A = 3$, so $A = -33$.

5

Integrate again

$f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{3/2} - 33x + B$.

6

Use P(4, 16)

$16 = \frac{2}{3}(64) + \frac{4}{3}(8) - 33(4) + B = \frac{160}{3} - 132 + B$.

7

Find B

$$B = \frac{284}{3}, \text{ so } f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{3/2} - 33x + \frac{284}{3}.$$

FINAL ANSWER

$$(a) y - 16 = -\frac{1}{3}(x - 4). \quad (b) f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{3/2} - 33x + \frac{284}{3}.$$

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10. A curve has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = ax - 12x^{\frac{1}{3}}$, where a is a constant
- $f''(x) = 0$ when $x = 27$
- the curve passes through the point $(1, -8)$

(a) find the value of a .

(3)

(b) Hence find $f(x)$.

(4)

(Total 7 marks)

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Question 29

WMA11/01 October 2021

T13 Integration

F03 Recovering from second-
derivative informationV02 Use a second-derivative
condition to find a parameter

1

Differentiate $f'(x)$

$$f'(x) = ax - 12x^{1/3}, \text{ so } f''(x) = a - 4x^{-2/3}.$$

2

Use $f''(27) = 0$

$$\text{Since } 27^{2/3} = 9, 0 = a - \frac{4}{9}, \text{ so } a = \frac{4}{9}.$$

3

Integrate $f'(x)$

$$\text{With } a = \frac{4}{9}, f'(x) = \frac{4}{9}x - 12x^{1/3}. \text{ Therefore } f(x) = \frac{2}{9}x^2 - 9x^{4/3} + c.$$

4

Use the point (1, -8)

$$-8 = \frac{2}{9} - 9 + c, \text{ so } c = -8 + 9 - \frac{2}{9} = \frac{7}{9}.$$

5

Write $f(x)$

$$f(x) = \frac{2}{9}x^2 - 9x^{4/3} + \frac{7}{9}.$$

FINAL ANSWER

$$(a) a = \frac{4}{9}. \quad (b) f(x) = \frac{2}{9}x^2 - 9x^{4/3} + \frac{7}{9}.$$

7. The curve C has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = \frac{2}{\sqrt{x}} + \frac{A}{x^2} + 3$, where A is a constant
- $f''(x) = 0$ when $x = 4$

(a) find the value of A .

(4)

Given also that

- $f(x) = 8\sqrt{3}$, when $x = 12$

(b) find $f(x)$, giving each term in simplest form.

(5)

(Total 9 marks)

Question 30

WMA11/01 May/June 2022

T13 Integration

F03 Recovering from second-
derivative informationV02 Use a second-derivative
condition to find a parameter

1

Differentiate $f'(x)$

$$f'(x) = 2x^{-1/2} + Ax^{-2} + 3, \text{ so } f''(x) = -x^{-3/2} - 2Ax^{-3}.$$

2

Use $f''(4) = 0$

$$-4^{-3/2} - 2A(4^{-3}) = 0, \text{ so } -\frac{1}{8} - \frac{A}{32} = 0 \text{ and } A = -4.$$

3

Substitute A into $f'(x)$

$$f'(x) = \frac{2}{\sqrt{x}} - \frac{4}{x^2} + 3.$$

4

Integrate to find $f(x)$

$$f(x) = 4\sqrt{x} + \frac{4}{x} + 3x + c.$$

5

Use $f(12) = 8\sqrt{3}$

$$f(12) = 8\sqrt{3} + \frac{1}{3} + 36 + c = 8\sqrt{3}, \text{ so } c = -\frac{109}{3}.$$

FINAL ANSWER

$$f(x) = 4\sqrt{x} + \frac{4}{x} + 3x - \frac{109}{3}.$$

10.

In this question you must show all stages of your working.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- the point $P(2, 8\sqrt{2})$ lies on C
- $f'(x) = 4\sqrt{x^3} + \frac{k}{x^2}$ where k is a constant
- $f''(x) = 0$ at P

(a) find the exact value of k ,

(4)

(b) find $f(x)$, giving your answer in simplest form.

(4)

(Total for Question 10 is 8 marks)

Question 31

WMA11/01 January 2024

T13 Integration

F03 Recovering from second-
derivative informationV02 Use a second-derivative
condition to find a parameter

1

Differentiate f prime

$$f'(x) = 4x^{3/2} + kx^{-2}, \text{ so } f''(x) = 6x^{1/2} - 2kx^{-3}.$$

2

Use $f''(P) = 0$

$$\text{At } P, x = 2 \text{ and } f''(2) = 0, \text{ so } 6\sqrt{2} - 2k(2^{-3}) = 0.$$

3

Find k

$$6\sqrt{2} - \frac{k}{4} = 0, \text{ hence } k = 24\sqrt{2}.$$

4

Integrate f prime

$$f(x) = \int (4x^{3/2} + 24\sqrt{2}x^{-2}) dx = \frac{8}{5}x^{5/2} - 24\sqrt{2}x^{-1} + c.$$

5

Use the point P

$$P(2, 8\sqrt{2}) \text{ lies on } C, \text{ so } 8\sqrt{2} = \frac{8}{5}(2^{5/2}) - \frac{24\sqrt{2}}{2} + c.$$

6

Find c

$$8\sqrt{2} = \frac{32\sqrt{2}}{5} - 12\sqrt{2} + c, \text{ so } c = \frac{68\sqrt{2}}{5}.$$

FINAL ANSWER

$$(a) k = 24\sqrt{2}. \quad (b) f(x) = \frac{8}{5}x^{5/2} - \frac{24\sqrt{2}}{x} + \frac{68\sqrt{2}}{5}.$$

Multi-constraint integration synthesis

WHAT THIS FAMILY ASKS

Resolve several independent constants or combine integration with an additional transformation or derivative demand.

VARIANT ROUTE

V01 Determine multiple coefficients before recovering f

2 Complete Classified questions

21 total marks

9. (i) Find

$$\int \frac{(3x + 2)^2}{4\sqrt{x}} dx \quad x > 0$$

giving your answer in simplest form.

(5)

(ii) A curve C has equation $y = f(x)$.

Given

- $f'(x) = x^2 + ax + b$ where a and b are constants
- the y intercept of C is -8
- the point $P(3, -2)$ lies on C
- the gradient of C at P is 2

find, in simplest form, $f(x)$.

(6)

(Total 11 marks)

Question 32

WMA11/01 January 2021

T13 Integration

F04 Multi-constraint integration synthesis

V01 Determine multiple coefficients before recovering f

1

Expand and simplify the integrand

$$\frac{(3x+2)^2}{4\sqrt{x}} = \frac{9x^2 + 12x + 4}{4x^{1/2}} = \frac{9}{4}x^{3/2} + 3x^{1/2} + x^{-1/2}.$$

2

Integrate term by term

$$\int \left(\frac{9}{4}x^{3/2} + 3x^{1/2} + x^{-1/2} \right) dx = \frac{9}{10}x^{5/2} + 2x^{3/2} + 2x^{1/2} + c.$$

3

Write in simplest form

$$\text{Factoring out } \sqrt{x} \text{ gives } \frac{\sqrt{x}(9x^2 + 20x + 20)}{10} + c.$$

4

Use the gradient condition at P

Since $f'(x) = x^2 + ax + b$ and the gradient at $P(3, -2)$ is 2, $f'(3) = 2$. Hence $9 + 3a + b = 2$, so $3a + b = -7$.

5

Integrate $f'(x)$

$$f(x) = \frac{x^3}{3} + \frac{a}{2}x^2 + bx + c. \text{ The y-intercept is } -8, \text{ so } c = -8.$$

6

Use $P(3, -2)$

$$-2 = \frac{27}{3} + \frac{9a}{2} + 3b - 8, \text{ so } \frac{9a}{2} + 3b = -3.$$

7

Solve for a and b

From $3a + b = -7$, $b = -7 - 3a$. Substitute into $\frac{9a}{2} + 3b = -3$ to get $a = -4$, and then $b = 5$.

8

Write $f(x)$ **Therefore** $f(x) = \frac{x^3}{3} - 2x^2 + 5x - 8.$ **FINAL ANSWER**

$$(i) \frac{\sqrt{x}(9x^2 + 20x + 20)}{10} + c. \quad (ii) f(x) = \frac{x^3}{3} - 2x^2 + 5x - 8.$$

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6.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- the point $P(4, -5)$ lies on C
- $f'(x) = \frac{2x^2 + ax + b}{4\sqrt{x}}$, where a and b are constants
- the gradient of the tangent to C at P is 7

(a) show that

$$4a + b = 24 \quad (2)$$

Given also that $a + b = -9$

(b) find, in simplest form, $f(x)$ (7)

Curve C is transformed to the curve with equation $y = f(x - 3)$

Given that point P is transformed to the point Q ,

(c) state the coordinates of Q . (1)

(Total for Question 6 is 10 marks)

Question 33

WMA11/01 January 2025

T13 Integration

F04 Multi-constraint integration synthesis

V01 Determine multiple coefficients before recovering f

1

Use the gradient at P

At $x = 4$, $f'(4) = 7$. Since $f'(x) = \frac{2x^2 + ax + b}{4\sqrt{x}}$, we get $\frac{32 + 4a + b}{8} = 7$.

2

Show the required equation

$32 + 4a + b = 56$, so $4a + b = 24$.

3

Solve for a and b

Together with $a + b = -9$, subtracting gives $3a = 33$, so $a = 11$ and $b = -20$.

4

Rewrite f prime

$$f'(x) = \frac{2x^2 + 11x - 20}{4\sqrt{x}} = \frac{1}{2}x^{3/2} + \frac{11}{4}x^{1/2} - 5x^{-1/2}.$$

5

Integrate

$$f(x) = \frac{1}{5}x^{5/2} + \frac{11}{6}x^{3/2} - 10x^{1/2} + c.$$

6

Use P to find c

$$-5 = \frac{1}{5}(4^{5/2}) + \frac{11}{6}(4^{3/2}) - 10(4^{1/2}) + c = \frac{16}{15} + c.$$

7

State f(x)

$$c = -\frac{91}{15}, \text{ so } f(x) = \frac{1}{5}x^{5/2} + \frac{11}{6}x^{3/2} - 10x^{1/2} - \frac{91}{15}.$$

8

Transform P

$y = f(x - 3)$ translates the curve 3 units to the right, so $P(4, -5)$ becomes $Q(7, -5)$.

FINAL ANSWER

$$(b) f(x) = \frac{1}{5}x^{5/2} + \frac{11}{6}x^{3/2} - 10x^{1/2} - \frac{91}{15}. \quad (c) Q = (7, -5).$$

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